TrialGPT
Matching patients to clinical trials with large language models

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Background

- **clinicaltrials.gov**
  - Public registry of clinical trials (U.S. National Library of Medicine)
  - Hundreds of thousands of studies worldwide
  - Trial information includes
    - condition / disease
    - intervention / treatment
    - recruitment status
    - study locations
    - **eligibility criteria**
  - Used to search and identify ongoing trials

- **Clinical Trial Eligibility Criteria**
  - Defines who can participate in a clinical trial
  - Composed of Inclusion & Exclusion criteria
  - All criteria must be satisfied for participation

Ex. Tirzepatide(Zepbound) Protocol

Motivation

- **Patient-Trial Matching**
  - Goal: identify clinical trials suitable for a patient
  - Challenges
    - Large number of trials
    - complex eligibility
    - Manual screening required

Example scenario:

✔ A Stage III lung cancer patient

✔ The patient did not respond to standard therapy

✔ The patient hopes to try experimental treatments

➡ **Participation in a clinical trial**

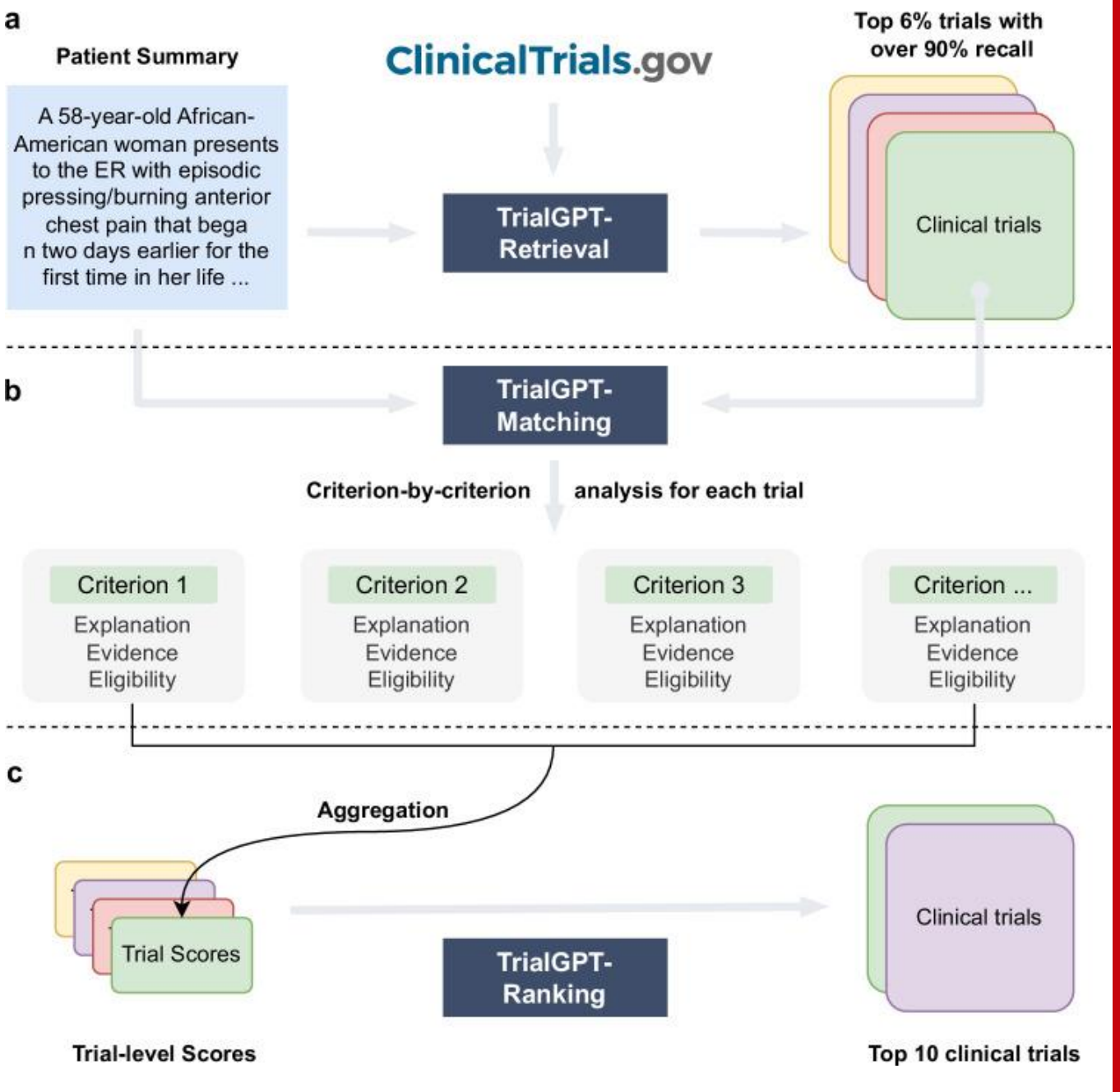
“Can we automate patient–trial matching?”

TrialGPT : Overview (Three-stage framework)

1. **Trial Retrieval**
   - retrieve candidate trials from a large database based on LLM-generated keywords

2. **Criterion-level Matching**
   - compare patient records vs eligibility criteria
   - determine whether each criterion is satisfied

3. **Trial Ranking**
   - aggregate criterion-level decisions
   - rank trials by relevance and eligibility



TrialGPT : (1) Trial Retrieval

Step 1. Keyword Generation

Patient Summary

A 58-year-old African-American woman presents to the ER with episodic pressing/burning anterior chest pain that began two days earlier for the first time in her life

…

LLMs

(GPT-4 or 3.5)

Keywords ranked by importance

(Chest Pain, Obesity, Hypertension, …)

$(w_1, w_2, \cdots, w_K), \quad K \leq 32$

Step 2. Hybrid Retrieval

- Search over J clinical trials: $t_j \ (j = 1, \ldots, J)$
- Use two retrieval methods: BM25 and MedCPT
- With keyword w_i, rank all clinical trials
  - $\mathrm{Rank}(\mathrm{BM25}, w_i, t_j) = P_{ij} \in [1, J]$
  - $\mathrm{Rank}(\mathrm{MedCPT}, w_i, t_j) = Q_{ij} \in [1, J]$

		BM25	MedCPT
w_1 = Chest Pain	Clinical Trial 1	P_{11}	Q_{11}
	Clinical Trial 2	P_{12}	Q_{12}
	Clinical Trial 3	P_{13}	Q_{13}
	⋮	⋮	⋮
	Clinical Trial J	P_{1J}	Q_{1J}
⋮			

TrialGPT : (1) Trial Retrieval

Feature	BM25 (Lexical Search)	MedCPT (Semantic Search)
Core Concept	Exact keyword matching	Contextual Embedding
Mechanism	Statistical scoring (TF-IDF based)	Neural model (biomedical transformer)
Search Logic	Does the exact term appear?	What is the underlying meaning?
Training	Unsupervised (Mathematical formula)	Contrastive learning on millions of PubMed logs
Strength	Precise for IDs, acronyms, and rare terms	Captures synonyms and clinical nuances
Limitation	Vocabulary Mismatch (e.g., fails on "Angina" vs "Chest Pain")	More computationally expensive than BM25

Step 3. Rank Fusion

- Fuse the ranks for the j-th clinical trial
  - $\mathrm{Rank}(\mathrm{BM25}, w_i, t_j) = P_{ij} \in [1, J]$
  - $\mathrm{Rank}(\mathrm{MedCPT}, w_i, t_j) = Q_{ij} \in [1, J]$

$$s_j = \sum_{i=1}^{K} \frac{1}{i} \times \left(\frac{1}{P_{ij} + C} + \frac{1}{Q_{ij} + C} \right) \quad (C = 20)$$

- Higher score when
  - Keyword importance is higher (smaller i)
  - Retrieved trial rank is higher (smaller rank)
- Retrieve $S(\leq J)$ clinical trials

TrialGPT : (2) Trial Matching

INPUT

- Patient note $(s_1, \ldots, s_p)$
- Trial background
  - title
  - condition
  - intervention
  - brief summary
- Eligibility criteria

chain-of-thought style prompting

✔ generate explanation first

✔ then predict evidence + label

OUTPUT

- Explanation
- Relevant sentence IDs
- Eligibility label : $E(\cdot)$
  - Included / Excluded
  - Not included / Not excluded
  - Not enough information
  - Not applicable

Patient note

The patient is a 64-year-old Asian male who was diagnosed with Stage II non-small cell lung cancer (NSCLC) last month. (s_1)
He has a medical history of hypertension, which is currently managed with … (s_2)

Eligibility criteria

Inc. 1: Age between 18 and 75 years
Inc. 2: Stage II or III pulmonary cancer
Exc. 1: History of other primary malignancies
Exc. 2: Severe or uncontrolled hypertension

Criterion	Explanation	Sentence ID	Label
Inc. 1	Age 64 satisfies requirement		Included
Inc. 2	Stage II NSCLC diagnosis		Included
Exc. 1	No other malignancy mentioned	NA	Not enough information
Exc. 2	Hypertension is controlled		Not Excluded

TrialGPT : (3) Trial Ranking

Linear Aggregation

- j-th clinical trial has
  - M inclusion criteria $(i_1, \ldots, i_M)$
  - N exclusion criteria $(e_1, \ldots, e_N)$

- 6 statistics summarizing trial eligibility (used as features for ranking trials)

$$\%\text{MetInc.} = \frac{\sum_{m=1}^{M} 1\{E(i_m) = \text{Included}\}}{M'}$$

$$\%\text{MetExc.} = \frac{\sum_{n=1}^{N} 1\{E(e_n) = \text{Excluded}\}}{N'}$$

$$\%\text{UnmetInc.} = \frac{\sum_{m=1}^{M} 1\{E(i_m) = \text{Not included}\}}{M'}$$

$$\%\text{UnmetExc.} = \frac{\sum_{n=1}^{N} 1\{E(e_n) = \text{Not excluded}\}}{N'}$$

$$\%\text{NoInc. Info} = \frac{\sum_{m=1}^{M} 1\{E(i_m) = \text{Not enough info.}\}}{M'}$$

$$\%\text{NoExc. Info} = \frac{\sum_{n=1}^{N} 1\{E(e_n) = \text{Not enough info.}\}}{N'}$$

$$M' = M - \sum_{m=1}^{M} 1\{E(i_m) = \text{Not applicable}\}$$

: # of applicable inclusion criteria

$$N' = N - \sum_{n=1}^{N} 1\{E(e_n) = \text{Not applicable}\}$$

: # of applicable exclusion criteria

TrialGPT : (3) Trial Ranking

❑ LLM Aggregation

- Prompt (Simplified)

You are a helpful assistant for clinical trial recruitment.
You will be given:
- a patient note
- a clinical trial description
- criterion-level eligibility predictions

Your task is to generate:
1. Relevance score R (0–100)
2. Eligibility score E (-R ~ R)
...
Please output a JSON dict:
{
"relevance_explanation": ...,
"relevance_score_R": ...,
"eligibility_explanation": ...,
"eligibility_score_E": ...
}

{
"relevance_explanation":
"The patient has stage II NSCLC, which matches the target condition of the clinical trial.",

"relevance_score_R": 82,

"eligibility_explanation":
"The patient satisfies the age criterion and disease stage, and there is no evidence of exclusion conditions.",

"eligibility_score_E": 65
}

- Relevance (R): How related the trial is to the patient
- Eligibility (E): How likely the patient can enroll

❑ **Feature Combination:** Linear + LLM Aggregation

$$\text{Score} = \%\text{MetInc.} - 1(\%\text{UnmetInc.} > 0) - 1(\%\text{MetExc.} > 0) + \frac{R}{100} + \frac{E}{100}$$

TrialGPT : Evaluation

Established Gold Standard

- 183 patient cases with 75,000+ expert-labeled patient–trial relevance annotations

Cohort	Source	Patients (N)	Avg. Notes Length	Labels per Patient
SIGIR 2016	ACM SIGIR	58	88.7 words	~66 labels
TREC 2021 CT	NIST TREC	75	156.2 words	~478 labels
TREC 2022 CT	NIST TREC	50	109.9 words	~708 labels
Total		**183**		**75,000+**

- SIGIR Special Interest Group on Information Retrieval
- ACM: Association for Computing Machinery
- TREC: Text Retrieval Conference
- NIST: National Institute of Standards and Technology

Evaluation Metrics

Metric	Focus	Clinical Importance
P@10	Quantity of Correct Hits	Minimizing "false alarms" in the top 10 list.
NDCG@10	Quality of Ranking Order	Ensuring the most eligible trials are at the very top.

- P@10: Precision at 10
- NDCG@10: Normalized Discounted Cumulative Gain at 10

TrialGPT : Results

Table 2 | Performance of different methods for ranking and excluding clinical trials

Application			Ranking		Excluding	Overall
Method / Metric	Baseline Method		NDCG@10	P@10	AUROC	Average
SciFive[46] (encoder-decoder)		Further trained on MedNLI[37]	0.4271	0.3787	0.5895	0.4652
BioBERT[47] (dual-encoder)		Further trained[48] on MNLI[49], SNLI[50], SciNLI[51], SciTail[52], MedNLI[37], and STSB[53]	0.4091	0.3746	0.5952	0.4596
PubMedBERT[54] (dual-encoder)			0.4327	0.3874	0.5976	0.4726
SapBERT[55] (dual-encoder)			0.4148	0.3738	0.5933	0.4606
BioLinkBERT[36] (cross-encoder)		Further trained on MedNLI[37]	0.4797	0.4281	0.6176	0.5085
TrialGPT-Ranking (GPT-3.5)		Feature combination	0.5395	0.5115	0.6582	0.5697
TrialGPT-Ranking (GPT-4)	Linear aggr.	Sign(% Included)	0.6097	0.5653	0.6765	0.6172
		Sign(% Not included)	0.4923	0.4556	0.6652	0.5377
		Sign(% Excluded)	0.3930	0.3715	0.6477	0.4707
		Sign(% Not excluded)	0.4130	0.3886	0.5820	0.4612
	LLM aggr.	Sign(Relevance)	0.7281	0.6700	0.7402	0.7128
		Sign(Eligibility)	0.7252	0.6724	0.7895	0.7290
	Feature combination		0.7275	0.6688	0.7979	0.7314

The Sign() function assigns suitable signs for the corresponding task, e.g., for "% Included", it will be "+" for ranking and "-" for excluding clinical trials. Aggr.: aggregation. NDCG@10: normalized discounted cumulative gain at 10. P@10: precision at 10.
AUROC the area under the receiver operating characteristic curve.

- TrialGPT significantly outperforms baseline models
- LLM aggregation performs better than linear aggregation
- Combining both signals yields the best result

TrialGPT : Results

Human Screening Study

- Two expert annotators: one with TrialGPT, one without
- 6 patient cases × 6 trials = 36 screening tasks

By patient

Case	1	2	3	4	5	6
Without TrialGPT (6 trials)	19.0 s	82.7 s	26.5 s	42.7 s	95.0 s	103.0 s
With TrialGPT (6 trials)	11.2 s	50.0 s	14.3 s	24.3 s	45.7 s	66.0 s
Time Saving	41.2%	39.5%	45.9%	43.0%	51.9%	35.9%

By Trial

Trial	A	B	C	D	E	F
Without TrialGPT (6 cases)	111.0 s	20.5 s	105.2 s	36.0 s	39.3 s	56.8 s
With TrialGPT (6 cases)	75.0 s	10.7 s	52.8 s	21.7 s	15.7 s	35.7 s
Time Saving	32.4%	48.0%	49.8%	39.8%	60.2%	37.2%

Total

	Short Cases	Long Cases	Annotator X	Annotator Y	All (36)
Without TrialGPT (18)	42.7 s	80.2 s	65.4 s	57.5 s	61.5 s
With TrialGPT (18)	25.2 s	45.3 s	37.8 s	32.7 s	35.3 s
Time Saving	41.0%	43.5%	42.2%	43.1%	42.6%
P-value	0.0182†	0.0003†	0.1633‡	0.2472‡	1.75e-5†

†: Paired t-test; ‡: independent t-test

TrialGPT : Conclusion

- LLMs can substantially improve patient–trial matching performance
- TrialGPT outperforms traditional biomedical retrieval models
- ~40% reduction in clinician screening time
- Best used as a clinician-assisting tool, not a fully automated system

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